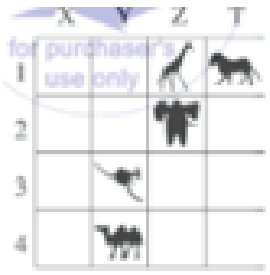
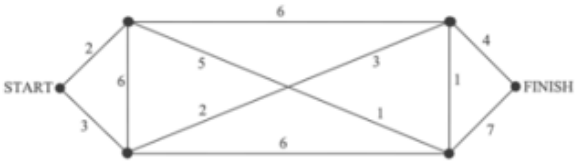
3 Points Each

1. Give the coordinates of the kangaroo.



(A) 2X
(B) 3Y
(C) 1Y
(D) 4Z
(E) 3T

2. A kangaroo is traveling from START to FINISH using the paths shown in the picture. Each segment is marked with the time (in minutes) which the kangaroo needs to travel that segment. What is the shortest time needed for the kangaroo to reach FINISH?



(A) 11 minutes
(B) 13 minutes
(C) 16 minutes
(D) 19 minutes
(E) 12 minutes

3. Among the puzzle pieces below, two have the same area. Which two?



(A) 4 and 2
(B) 1 and 5
(C) 1 and 3
(D) 4 and 5
(E) 3 and 5

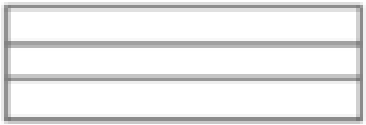
4. What number is the smallest natural number greater than 360 and at the same time a square of a natural number?

(A) 400
(B) 362
(C) 361
(D) 900
(E) other number

5. One night and day period on Mars is 40 minutes longer than on Earth. How much longer is a week on Mars than on Earth?

(A) 4 h 40 min
(B) 2 h 80 min
(C) 7 h 20 min
(D) 40 min
(E) 0 min

6. How many rectangles are there in the picture?



(A) 1
(B) 3
(C) 4
(D) 5
(E) 6

7. A wall clock strikes at every hour (the number of strikes corresponds to the time, so example, at 10 a.m. and at 10 p.m. you will hear 10 strikes). The clock also strikes once at the half-hour mark. How many strikes can be heard in one 24-hour period?

(A) 24
(B) 136
(C) 180
(D) 196
(E) 240

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8. It is now the spring of 1998. The last Summer Olympics took place in 1996, and the last Winter Olympics finished just a few weeks ago. Both the Summer and Winter Olympics take place every 4 years. Counting both the summer and winter competitions, how many more times will the Olympics take place before March 20, 2051?

(A) 13
(B) 16
(C) 25
(D) 26
(E) other answer

9. In how many ways can you place two identical 1 dollar coins in three pockets?

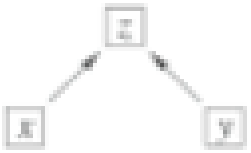
(A) 2
(B) 3
(C) 4
(D) 6
(E) 8

10. Andy is wearing a t-shirt with the word KANGUR on it. He is standing in front of a mirror. What word does he see when he looks at his t-shirt in the mirror?

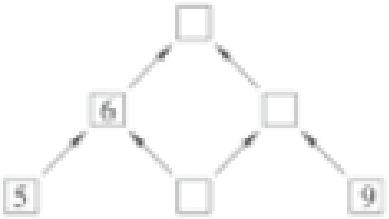
(A) KANGUR (B) RUGNAK (C) RUGNAK
(D) KANGUR (E) RUGNAK

4 Points Each

11. What number is at the top of the pyramid if it is formed according to the pattern shown below?



$$z = \frac{x+y}{2}$$



(A) 5
(B) 7
(C) 8
(D) 9
(E) 12

12. A watermelon weighs $\frac{4}{5}$ kg more than $\frac{4}{5}$ of the same watermelon. How much does the watermelon weigh?

(A) $\frac{8}{5}$ kg
(B) 4 kg
(C) 3 kg
(D) 4.5 kg
(E) 5 kg

13. There are stools and chairs in the room. Each stool has 3 legs and each chair has 4 legs. Altogether there are 17 legs. How many chairs are there in the room?

(A) 5
(B) 4
(C) 3
(D) 2
(E) 1

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14. If □ + ○ = 30, □ + △ + △ = 160, and △ + ○ = 80, then □ + △ + ○ + ○ = ?

(A) 80
(B) 100
(C) 110
(D) 210
(E) 90

15. When from any three-digit number we subtract that number written backwards, the difference will always be a number that is divisible by:

(A) 7
(B) 2
(C) 5
(D) 9
(E) 13

16. When Mr. Kowalski was asked how old he was, he said, "I have lived 44 years, 44 months, 44 weeks, 44 days, and 44 hours." How many years old is Mr. Kowalski?

(A) 44
(B) 47
(C) 48
(D) 49
(E) 50

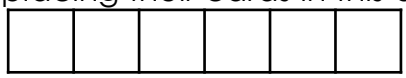
17. There are 3 married couples. In how many ways can we form a three-person group in which there will not be a married couple?

(A) 1
(B) 2
(C) 6
(D) 8
(E) 20

18. On Monday morning, a snail fell down a well which is 10 meters deep. During the day, it climbs up 2 meters, and during the night it slides down 1 meter. On what day of the week will the snail get out of the well?

(A) Tuesday
(B) Thursday
(C) Saturday
(D) Sunday
(E) Monday

19. John and Stan each have three cards marked with digits. John's cards are marked with the digits 2, 4, and 6, and Stan's cards are marked with the digits 1, 3, and 5. They are taking turns placing their cards in this diagram:



John will fill in the first spot on the left, Stan the second spot, and so on. John is trying to make the final number as small as possible, and Stan is trying to make it as large as possible. What number will they form?

(A) 123456
(B) 654321
(C) 254361
(D) 253146
(E) 253416

20. At a fair, the tickets for four various types of rides cost 2 dollars, 3 dollars, 4 dollars, and 5 dollars, respectively. A class took a field trip to the fair. They bought enough tickets for each student to go on each of the four rides once. The tickets cost 280 dollars altogether. How many tickets did they buy?

(A) 14
(B) 20
(C) 40
(D) 80
(E) 140

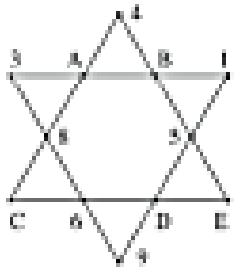
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5 Points Each

21. A juice carton which is $\frac{3}{4}$ full contains enough juice to fill 1 $\frac{1}{2}$ glasses. How many glasses will the juice from 5 full cartons fill?

(A) $7\frac{1}{2}$
(B) $3\frac{3}{4}$
(C) 8
(D) 10
(E) $8\frac{1}{4}$

22. Whole numbers from 1 to 12 are placed in the figure in such a way that the sum of the four numbers found along each segment is the same (see the picture). Under which letter ı?

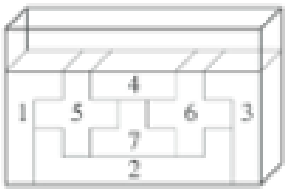


(A) A
(B) B
(C) C
(D) D
(E) E

23. What is the ones digit in the number $1^2+2^2+3^2+4^2+5^2+6^2+7^2+8^2+9^2+10^2$?

(A) 1
(B) 3
(C) 5
(D) 7
(E) 9

24. Choose the order in which it would be impossible to place the blocks in the box.



(A) 2, 7, 5, 6, 4, 1,3
(D) 2, 7, 6, 5, 3, 1,4
(B) 2, 7, 5, 1, 6, 4,3
(C) 2, 7, 6, 3, 4, 5, 1
(E) 2, 7, 5, 1, 6, 3, 4

25. In the diagram below, house X is shown 4 times (from different sides), and house Y is shown just once. Which picture shows house Y?

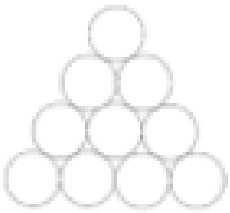


26. Four teams took part in a soccer tournament. The rules were:
a) each team plays against each of the other teams exactly once, and
b) a team gets 3 points for winning, 0 points for losing, and 1 point if there is a tie.
At the end of the tournament, the teams had 5 points, 3 points, 3 points, and 2 points, respectively. How many of the games ended in a tie?

(A) 1
(B) 2
(C) 3
(D) 4
(E) 5

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27. Ten coins were placed as shown in the picture. What is the smallest number of coins that need to be removed so that an equilateral triangle cannot be formed by the centers of any three of the remaining coins?



(A) 3
(B) 4
(C) 5
(D) 6
(E) 7

28. Snow White lined up the Seven Dwarfs from shortest to tallest. She divided 77 berries among them. The shortest dwarf got a certain number of berries, the next one got one berry more, and so on. How many berries did the tallest dwarf get?

(A) 17
(B) 8
(C) 14
(D) 10
(E) 15

29. In the basketball semi-finals, team A plays against team B and team C plays against team D. The winners of these two games will play against each other for first and second place, and the losers will play for third and fourth place. How many possible outcomes are there?

(A) 4
(B) 8
(C) 12
(D) 16
(E) 24

30. The area of the shaded triangle is one-fourth of the area of the rectangle (see the picture). What part of the length of the base of the rectangle is the base of the triangle?



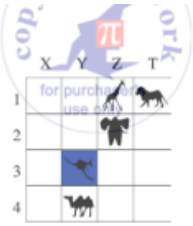
(A) $\frac{1}{2}$
(B) $\frac{1}{3}$
(C) $\frac{1}{4}$
(D) $\frac{3}{4}$
(E) $\frac{2}{3}$

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Answers key:

1. B
2. A
3. B
4. C
5. A
6. E
7. C
8. D
9. D
10. C
11. B
12. B
13. D
14. C
15. D
16. C
17. D
18. A
19. C
20. D
21. D
22. E
23. C
24. C
25. B
26. E
27. B
28. C
29. D
30. A

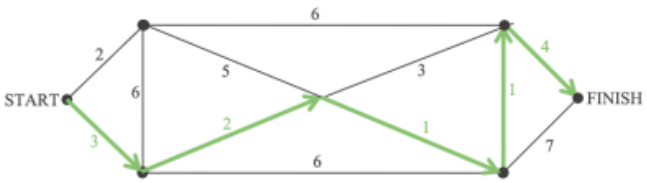
Solutions for Year 1998



1. (B) 3Y

The kangaroo is in the 3rd row and the Y column, so its coordinates are 3Y.

2. (A) 11 minutes



Try different paths. The quickest path is marked by green arrows and the time along this path is 3 + 2 + 1 + 1 + 4, which is 11 minutes.

3. (B) 1 and 5

The puzzle pieces are formed by identical squares with identical rounded tabs (shown in blue) and cut-outs. Each cut-out removes the area equivalent to one tab.



For each puzzle piece, the area is shown underneath. S is the area of one square and T is the area of one tab. Only pieces 1 and 5 have the same area. Notice that piece 5 is simply piece 1 rotated 90° clockwise.

4. (C) 361

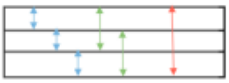
$16^2 = 256$, $17^2 = 289$, $18^2 = 324$, $19^2 = 361$, and $20^2 = 400$ are consecutive perfect squares, so 361 is the smallest natural number greater than 360 that is a square of a natural number.

5. (A) 4 h 40 min

One week on Mars is 7×40 min = 280 min longer than one week on Earth. $280 = 4 \times 60 + 40$, so 280 min = 4 h 40 min. One week on Mars is 4 h 40 min longer than one week on Earth.

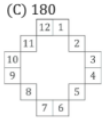
6. (E) 6

The rectangles are identified by their heights. There are 6 of them.



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7. (C) 180



The number of hourly strikes for half a day is shown to the left. In each row the sum of numbers is 13, so the number of all hourly strikes is $6 \times 13 = 78$ for half a day. For one whole day the number of hourly strikes is $2 \times 78 = 156$ and there are also 24 strikes at the half-hour marks, so $156 + 24 = 180$ strikes can be heard in one 24-hour period.

8. (D) 26

Following spring 1998, the Summer Olympics will take place in the years 2000, 2004, 2008, 2012, 2016, 2020, 2024, 2028, 2032, 2036, 2040, 2044, and 2048. The Winter Olympics will take place in the years 2002, 2006, 2010, 2014, 2018, 2022, 2026, 2030, 2034, 2038, 2042, 2046, and 2050. Counting both the summer and winter competitions, the Olympics will take place 26 times between spring 1998 and March 20, 2051.

9. (D) 6

There are 3 ways to keep one pocket empty and 3 ways to keep two pockets empty.



10. (C) ЯUƃИAЖ

KANGUR | ЯUƃИAЖ

The mirror image of KANGUR is the same as a symmetric image of KANGUR with respect to a line perpendicular to the line of KANGUR and it is shown to the right. Symmetry with respect to a line parallel to the line of KANGUR gives us (A) ꓘ⅄ИGꓵꓤ. Rotation of KANGUR by 180° gives us (E) ЯꓵƃNⅤꓘ. KANGUR written from right to left is (B) RUGNAK and its mirror image is (D) ЖАИǤUЯ.

11. (B) 7



Place 7 (shown in red) in the empty space in the bottom row since $\frac{5+7}{2} = (5+7) \div 2 = 12 \div 2 = 6$. Place 8 above 7 and 9 because $\frac{7+9}{2} = 16 \div 2 = 8$. Then, place 7 above 6 and 8 as $\frac{6+8}{2} = 14 \div 2 = 7$. Thus, the number at the top of the pyramid is 7.

12. (B) 4 kg

The difference between weights of 1 watermelon and $\frac{4}{5}$ of the same watermelon is $\frac{1}{5}$ of the weight of the watermelon and that difference is $\frac{4}{5}$ kg. Hence, $\frac{1}{5}$ of the watermelon weighs $\frac{4}{5}$ kg, so one whole watermelon weighs $5 \times \frac{4}{5}$ kg $= 4$ kg.

---

13. (D) 2

17 is neither a multiple of 4 nor a multiple of 3, so there is at least one pair of a chair and a stool. This pair accounts for 7 legs. $17 - 7 = 10$ and again 10 is neither a multiple of 4 nor a multiple of 3, so there is another pair of a chair and a stool. These two pairs account for $2 \times 7 = 14$ legs. The other $17 - 14 = 3$ legs belong to one stool, so there are 3 stools and 2 chairs in the room. $3 \times 3 + 2 \times 4 = 17$ is the number of all legs.

14. (C) 110

$\square + \triangle + \bigcirc + \bigcirc = (\square + \bigcirc) + (\triangle + \bigcirc) = 30 + 80 = 110$.

We do not need the middle equation, $\square + \triangle + \triangle = 160$, for the calculation. Also, we don't actually need to figure out what number each shape represents.

15. (D) 9

The difference $(100x + 10y + z) - (100z + 10y + x)$ simplifies to $99x - 99z$ or $99 \times (x - z)$, so the difference is always a multiple of 99, 33, 11, 9, 3, and 1. Only 9 is among the five given options, so 9 is the answer.

16. (C) 48

44 weeks + 44 days + 44 hours = 44 weeks + 7 weeks + 2 days + 1 day + 20 hours, so 44 weeks + 44 days + 44 hours = 51 weeks + 3 days + 20 hours, which is between 51½ and 52 weeks.

One month is between 4 weeks (28 days) and 4½ weeks (31½ days), so 44 months are between $44 \times 4 = 176$ and $44 \times 4\frac{1}{2} = 176 + 22 = 198$ weeks. Therefore, all of Mr. Kowalski's months, weeks, days, and hours are between $51\frac{1}{2} + 176 = 227\frac{1}{2}$ and $52 + 198 = 250$ weeks.

1 year is less than 52½ weeks (367½ days), so 4 years is less than $4 \times 52\frac{1}{2} = 210$ weeks. $210 < 227\frac{1}{2}$, so 4 years is less than all of Mr. Kowalski's months, weeks, days, and hours.

1 year is more than 52 weeks (364 days), so 5 years is more than $5 \times 52 = 260$ weeks. $260 > 250$, so 5 years is more than all of Mr. Kowalski's months, weeks, days, and hours.

In summary, Mr. Kowalski's has lived between $4 + 44 = 48$ years and $5 + 44 = 49$ years. Therefore, Mr. Kowalski is 48 years old.

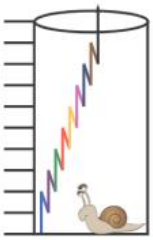
17. (D) 8

Let A_1, B_1 be the first couple. Similarly, A_2, B_2 and A_3, B_3 be the 2nd and 3rd couple, respectively. Then the 3-person groups where there will not be a married couple are: $\{A_1,A_2,A_3\}$, $\{A_1,A_2,B_3\}$, $\{A_1,B_2,A_3\}$, $\{A_1,B_2,B_3\}$, $\{B_1,A_2,A_3\}$, $\{B_1,A_2,B_3\}$, $\{B_1,B_2,A_3\}$, and $\{B_1,B_2,B_3\}$. There are 8 such groups.

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18. (A) Tuesday

Tuesday	2 up and out
Monday	2 up 1 down = 1 up
Sunday	2 up 1 down = 1 up
Saturday	2 up 1 down = 1 up
Friday	2 up 1 down = 1 up
Thursday	2 up 1 down = 1 up
Wednesday	2 up 1 down = 1 up
Tuesday	2 up 1 down = 1 up
Monday	2 up 1 down = 1 up



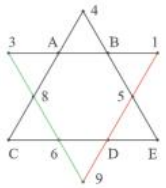
On the 9th day the snail climbs up from the level of 8 meters above the bottom of the well to the level of 10 meters from the bottom and gets out of the well.
This happens on Tuesday.

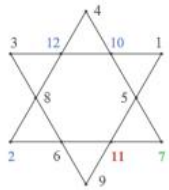
19. (C) 254361
The smallest 3-digit number John can create is 2 4 6 and Stan's largest 3-digit number is 5 3 1. Put them together in alternating order (John first) to see 254361 as the number formed by both of them.

20. (D) 80
The class spent \$2 + \$3 + \$4 + \$5 = \$14 per student to buy 4 different tickets for each student. For the whole class they spent \$280, so there were 280 ÷ 14 = 20 students in the class. Since the school bought 4 tickets for each student, they bought 20 × 4 = 80 tickets.

21. (D) 10
$\frac{3}{4}$ of the capacity of one carton is the same as $\frac{3}{2}$ of the capacity of one glass, so $\frac{1}{4}$ of the capacity of one carton is the same as $\frac{1}{2}$ of the capacity of one glass, or the capacity of 1 cartoon is the same as the capacity of 2 glasses since $4 \times \frac{1}{4} = 1$ and $4 \times \frac{1}{2} = 2$. Therefore, juice from 5 cartons will fill 10 glasses.

22. (E) E

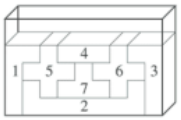




The numbers along the green segment add up to 3 + 8 + 6 + 9 = 26, so 26 is the sum of numbers along any segment. Hence, the number under D is 26 – (1 + 5 + 9) = 11. Also, the sum of numbers under C and E is 9 since 26 – (6 + 11) = 9. The numbers still available are 2, 7, 10, and 12. To get 9 as the sum we can only use 2 and 7. Look at the segment with E and B. If the number under E were 2, then the number under B would be 26 – (2 + 5 + 4) = 15; however, 15 is not among numbers from 1 to 12.
Thus, 7 is under E and we can find all the other numbers. The results are shown above.

23. (C) 5
The first ten perfect squares are 1, 4, 9, 16, 25, 36, 49, 64, 81, and 100. The sum of ones digits of these squares is 1 + 4 + 9 + 6 + 5 + 6 + 9 + 4 + 1 + 0 = 45, so the ones digit of the sum $1^2 + 2^2 + 3^2 + 4^2 + 5^2 + 6^2 + 7^2 + 8^2 + 9^2 + 10^2$ is 5.

24. (C) 2, 7, 6, 3, 4, 5, 1



For orders (A) 2, 7, 5, 6, 4, 1, 3; (B) 2, 7, 5, 1, 6, 4, 3; (D) 2, 7, 6, 5, 3, 1, 4; and (E) 2, 7, 5, 1, 6, 3, 4 we can put all the blocs in the places they are shown in the box to the left. For (C), the blocks 2, 7, 6, and 3 can be easily placed in the box in that order. After that, placing block 4 before block 5 would not allow block 5 to fill the remaining empty spaces.

25. (B)



In (B) when we face the door of the house, the long window is on the right side.

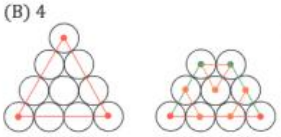
(A)  (C)  (D)  (E) 

In (A) there are two windows on the right side, so the long window (not visible) is on the left side as shown in (C). The pictures of the house in (D) and (E) are consistent with (A) and (C), so the house Y is shown in (B).

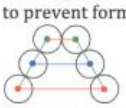
26. (E) 5
There were four teams: team A earned 5 points, team B earned 3 points, team C earned 3 points, and team D earned 2 points. The four teams played 6 games: {A,B}, {A,C}, {A,D}, {B,C}, {B,D}, and {C,D}.
Each game yields either 3 + 0 = 3 points or 1 + 1 = 2 points. The total of all points is 5 + 3 + 3 + 2 = 13. Neither 3 nor 2 are factors of 13, so at least one game yielded 3 points and at least one game yielded 2 points. The other 13 – (3 + 2) = 8 points are distributed among 4 games. 8 as a sum of 4 numbers, each number being either 3 or 2, can be written only in one way: 8 = 2 + 2 + 2 + 2. Thus, 5 games ended in a tie.
There is only one distribution of points. A and B were tied, A and C were tied, and A beat D. The other three games were tied.

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27. (B) 4



One of the corner coins with red centers must be removed. Suppose it is the top coin. The triangle with three orange sides and two triangles each with 3 sides of different colors have the same size but they do not share vertices. From each of these three triangles we have to remove at least one coin to prevent forming equilateral triangles. Hence, at least $1 + 3 = 4$ coins must be removed to prevent formation of equilateral triangles. It can be done as shown below to the left.



If d is a diameter of one coin, then d is the length of the orange segment, $2d$ is the length of the blue segment and $3d$ is the length of the red segment. The figure is symmetric with respect to the vertical line, so to see all triangles of different sizes pick 2 vertices from the right green segment (it can be done in 3 ways) and one vertex from the left green segment (there are 3 options), so $3 \times 3 = 9$ is the number of triangles of different sizes. For each of these triangles you can easily see that the three sides are not of the same length or not every of the three angles has 60°, so none of the triangles are equilateral.

A word of caution: Not every way of removing four coins guarantees that no equilateral triangles can be formed. For example, removing all corner coins with red centers would not work, as shown in the picture with the removed coins colored white.



28. (C) 14

If Snow White divided 77 berries among 7 dwarfs equally, then each dwarf would get $77 \div 7 = 11$ berries. When lined up from shortest to tallest, the middle dwarf has 3 shorter dwarfs on one side and 3 taller dwarfs on his other side. The middle dwarf has 11 berries. His shorter neighbor has one berry less than the middle dwarf, and the taller neighbor has one more, so one has 10 berries and the other has 12 berries. The shorter neighbor of the dwarf with the 10 berries has two berries less than the middle dwarf and the taller neighbor of the dwarf with 12 berries has two more, so one has 9 and the other 13. The shortest dwarf has three berries less than the middle dwarf and the tallest dwarf has three more. Thus, the distribution of the 77 berries from the shortest to the tallest dwarf is 8, 9, 10, 11, 12, 13, **14**. Hence, the tallest dwarf got 14 berries.

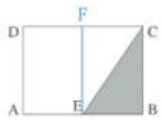
29. (D) 16

Any of the 4 teams can win the competition. Once the winner is selected, any of the 2 teams from the other matchup can be the second-place team. If the first and second places are known, then either one from the other 2 teams can be the third-placed team, so the number of all possible outcomes is $4 \times 2 \times 2 = 16$.

Here are all 16 outcomes: ACBD, ACDB, ADBC, ADCB; BCAD, BCDA, BDAC, BDCA; CADB, CABD, CBDA, CBAD; DACB, DABC, DBCA, DBAC.

Here is another way to find the possible outcomes: If A is the winner, then B can be only third or fourth and either C or D is second. Assuming that A places first in this competition, the possible outcomes are: ACBD, ACDB, ADBC and ADCB. There are 4 possible outcomes in the case when A is the winner. The same argument works when B, C, or D is the winner of the whole competition, so there are $4 \times 4 = 16$ possible outcomes.

30. (A) $\frac{1}{2}$



Draw a segment EF which is perpendicular to AB, so the triangles EBC and CFE are equal. Hence, the area of rectangle EBCF is $\frac{1}{4} + \frac{1}{4} = \frac{1}{2}$ of the area of rectangle ABCD. Thus, E is the midpoint of the segment AB, so the base of the shaded triangle is $\frac{1}{2}$ of the length of the base of the rectangle.

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